

Interview Questions — Python - Fundamentals of python language

Total Questions: 57

1. What is Python, and why is it considered a popular programming language?

Hint: Think about Python's readability, community, and typical uses.

2. Give two real-world use cases where Python is commonly used.

3. How do you install Python on your computer and write your first script in VS Code?

Hint: Mention both the installation process and the steps to create and run a .py file.

4. Explain how Python code is executed from writing to output.

5. What are variables in Python, and how do you declare one?

Hint: Think about assignment and naming.

6. List two rules you must follow when naming variables in Python.

7. What is the difference between primitive and non-primitive data types in Python?

Hint: Focus on built-in simple types vs. more complex structures.

8. Write Python code to create variables for a student profile containing name, age, and enrolled course.

9. What does the type() function return in Python? Give an example.

10. How do you get input from a user in Python, and how do you display output?

Hint: Mention the relevant built-in functions.

11. How would you convert a string input to an integer in Python?

12. Write a Python program that takes two numbers from the user and prints their sum.

Hint: Remember to convert input to the correct type.

13. What is an escape character in Python strings? Give an example.

14. Explain the difference between if, elif, and else statements in Python.

15. Write a Python program that checks if a number entered by the user is positive, negative, or zero.

16. Describe how you would use nested if statements. Give a simple example.

17. Write a Python program to determine if someone is eligible to donate blood. (Eligibility: age $\geq$ 18 and weight $\geq$ 50kg)

18. What is a for loop in Python used for?

19. Write a for loop to print each fruit name from a list: ['apple', 'banana', 'cherry'].

20. How can you use a for loop to count the number of vowels in a string?

21. What is the difference between a for loop and a while loop in Python?

Hint: Consider how each loop determines when to stop.

22. Write a while loop to print numbers from 1 to 5.

23. How would you create a simple star pattern like this using a while loop? * * * * *

24. What happens if the condition in a while loop never becomes False?

Hint: Think about infinite loops.

25. Explain the use of break, continue, and pass statements in loops.

26. Write a for loop that prints each fruit in the list ['apple', 'banana', 'cherry'], but skips 'banana'.

27. How would you use break to exit a loop when a specific value is found?

28. Give an example where pass would be useful in a Python loop or function.

29. How can you access the first and last character of a string in Python?

Hint: Think about string indexing.

30. Write Python code to reverse a string entered by the user.

31. What is string slicing? Give an example to extract a substring.

32. Name two string methods in Python and explain what they do.

33. How do you create a list in Python and add a new item to it?

34. Write code to remove the third element from a list.

35. Explain the difference between the `append()` and `insert()` methods for lists.

36. How would you sort a list of numbers in ascending order?

37. What is a tuple in Python, and how is it different from a list?

Hint: Focus on mutability.

38. Write code to create a tuple with three elements and access its second value.

39. How can you concatenate two tuples in Python?

40. What happens if you try to change an element of a tuple after creation?

41. What is a dictionary in Python, and how do you add a new key-value pair?

42. Write code to update the value of an existing key in a dictionary.

43. How can you get all the keys and all the values from a dictionary?

44. Explain the difference between `keys()`, `values()`, and `items()` methods in dictionaries.

45. How do you define a function in Python that takes two numbers and returns their product?

46. What is a lambda function? Write a lambda to square a number.

47. How would you use `map()` to double every number in a list?

48. Give an example of using `filter()` with a lambda to get only even numbers from a list.

49. How would you explain the concept of return values in Python functions?

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50. Write a function that takes a list of numbers and returns the maximum value.
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51. How would you use `reduce()` to calculate the sum of all elements in a list?
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52. When would you use a function instead of repeating code in your program?
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53. How could you use ChatGPT or GitHub Copilot to help you write or debug Python code, and what steps would you take to ensure the code it suggests is correct?
- Hint:** Think about testing and understanding suggestions.
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54. If you need to quickly generate example data or boilerplate code for your Python project, how might AI tools help, and what would you double-check before using their output?
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55. Suppose you need to process a list of student records and extract names of students who scored above 80. Which combination of data structures and control flow statements from this module would you use, and why?
- Hint:** Consider lists, for loops, and conditionals together.
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56. How would you design a Python function that takes a string, checks if it is a palindrome, and returns `True` or `False`? Mention which string methods and control flow you would use.
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57. Imagine you are building a student management system. Which data type would you use to store each student's profile (name, age, courses), and how would you access or update their information efficiently?
- Hint:** Think about dictionaries and their key-value access.
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